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Investigating Strong Force with Machine Learning

Kate Vinit Vijay (2020PH10704)
Paras Garg (2020PH10715)

Adviser: Prof. Tobias Toll

Abstract: We aim to study parton distribution function (gluon and quark distribution) using Inclusive Deep Inelastic Scattering in the low x-region. Currently there are multiple models such as the b-CGC, b-Sat, etc. which are used to probe the proton structure using Inclusive or Exclusive diffraction. We want to train a Machine Learning model on data provided by HERA measurements of the inclusive DIS process. As there is less data available for the exclusive processes, we considered using the inclusive DIS. Lastly, we aim to draw inferences by comparing the results of our trained model and the existing physics models.

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1 The Strong Force

The strong force is termed a fundamental force because it's one of the four fundamental forces in nature that govern the interactions between particles at the most basic level. These four forces—gravity, electromagnetism, the weak nuclear force, and the strong nuclear force—form the basis for all interactions in the universe.[12]

The strong force specifically acts between quarks, which are the building blocks of particles like protons and neutrons. It's responsible for holding quarks together to form protons, neutrons, and other particles. It is termed strong because it's the most powerful of the fundamental forces, but it is confined within the atomic nucleus, acting at very short ranges, making it crucial for understanding the structure of matter at a fundamental level.

Quarks and gluons are fundamental particles that are key constituents of matter and play a fundamental role in the Standard Model of particle physics.[13]

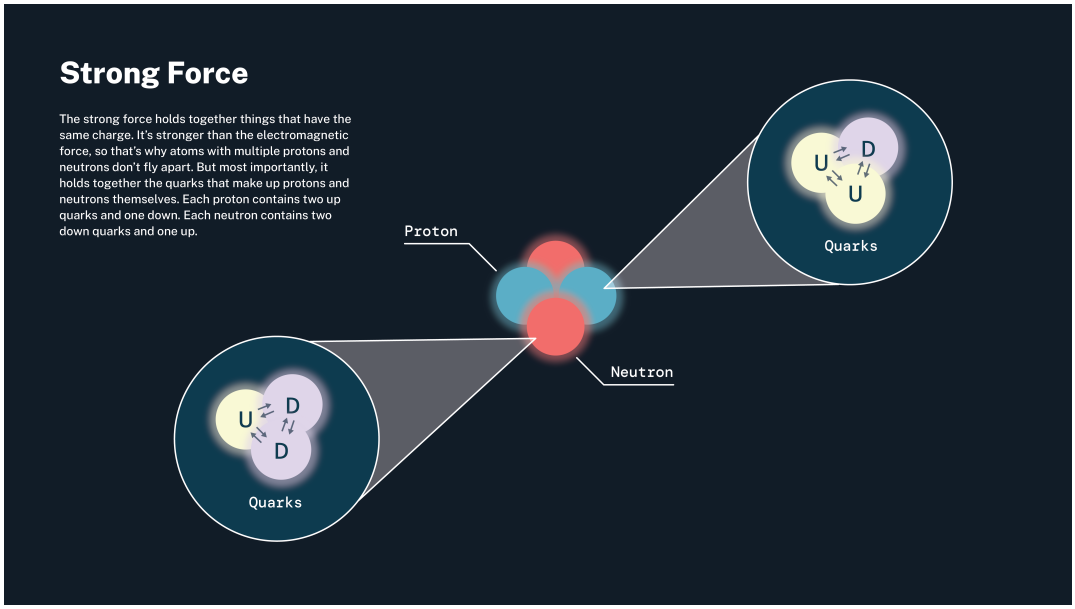


Figure 1: Strong Force inside nucleus [6]

- Quarks:
  - **Types of Quarks:** Quarks come in six types or flavors: up, down, charm, strange, top, and bottom.
  - **Charge and Mass:** Quarks carry fractional electric charges: +2/3 or -1/3 of the elementary charge (the charge carried by an electron). They also have mass, though they are confined within particles like protons and neutrons and are never found in isolation due to a phenomenon known as color confinement.
  - **Color Charge:** Quarks also carry a property called "color charge," but this is not related to colors as we perceive them. Instead, it's a quantum property analogous to color in the way it combines and interacts. Quarks come in three "colors": red, green, and blue.
  - **Confinement:** Quarks are always found in combinations that result in color-neutral particles due to a phenomenon called confinement. For example, a proton consists of two up quarks (each with a charge of +2/3) and one down quark (with a charge of -1/3), resulting in a net charge of +1.

- Gluons:
  - **Gluons as Force Carriers:** Gluons are the carrier particles of the strong force, which binds quarks together within particles like protons and neutrons.
  - **Interaction:** Unlike photons (the carriers of the electromagnetic force), gluons carry the color charge themselves, allowing them to interact with quarks and with other gluons.
  - **Self-Interaction:** One of the unique aspects of the strong force is that gluons can interact with other gluons, leading to the concept of gluon self-interaction, which makes the strong force stronger as quarks move farther apart (as opposed to the weakening of the electromagnetic force as charged particles move apart).
- Quark and Gluon Interactions:
  - **Strong Force:** Quarks interact with each other through the exchange of gluons. These interactions are described by quantum chromodynamics (QCD), the theory governing the strong force.
  - **Binding in Hadrons:** The strong force mediated by gluons binds quarks together to form composite particles called hadrons, like protons and neutrons.
  - **Asymptotic Freedom:** At very high energies or short distances, quarks and gluons behave almost as free particles due to a property called asymptotic freedom, a key feature of QCD.[12]

2 Previous efforts to understand the Strong Force

Deep Inelastic scattering is a technique that is used extensively to study the cross sections of interactions between an incoming electron beam and the scattered hadrons. This technique has been used extensively, historically, as probes to study the internal structures of protons. Diffractive DIS technique entails a process where the proton states intact, mediated only by a color-neutral exchange such as gluon-gluon exchange. Two methods are used:

- **Inclusive Deep Inelastic Scattering:** Inclusive DIS involves measuring the overall characteristics of the scattered particles, focusing on quantities like the total scattering cross-section or the energy distribution of the scattered particles. Here, the goal is to understand the average behavior of the partons (quarks and gluons) inside the nucleon without specifically identifying the individual particles involved in the interaction.
  By analyzing the energy and momentum transfer between the incident lepton and the nucleon, we can extract information about the parton distribution functions (PDFs). These functions describe the probability of finding a particular parton carrying a fraction of the nucleon's momentum at a given momentum scale. Inclusive DIS provides crucial insights into the overall structure and behavior of partons within nucleons but doesn't provide detailed information about specific parton configurations.[10]

$$e + p \rightarrow \ anything$$

- **Exclusive Deep Inelastic Scattering:** exclusive DIS focuses on identifying and studying the specific particles produced in the scattering process. This method aims to analyze and understand the internal structure of the nucleon by observing the specific final-state particles resulting from the collision between the lepton and the nucleon. By measuring the momenta, energies, and angles of the particles produced in the collision, researchers can reconstruct the kinematics of the interaction. Exclusive DIS allows for a more detailed investigation of the internal structure of partons, providing information about their spatial distributions, momentum distributions, and the nature of their interactions.[11]

$$e + p \rightarrow \ specific\ final\ state$$

Multiple models have been proposed in the literature that try to explain the data obtained from HERA experiments of the interaction between incoming electron beams and the nucleus.

- **GBW Model:** The dipole model became an important tool in investigations of deep-inelastic scattering due to the initial observation of Golec-Biernat and Wusthoff (**GBW**) that a simple ansatz for the dipole cross section integrated over the impact parameter b, $\sigma_{q\bar{q}}$, was able to describe simultaneously the total inclusive and diffractive DIS cross sections:

$$\sigma_{q\bar{q}}^{GBW}(x,r) = \sigma_0(1 - e^{\frac{-r^2 Q_s^2(x)}{4}})$$

  where σ_0 is a constant, x is the Bjorken constant, r is the dipole size, and $Q_s(x)$ denotes the x dependent saturation scale.[1]

- **BGBK Model**: The assumption of dipole saturation provided an attractive theoretical background for the investigation of the transition from the perturbative to non-perturbative regimes in the HERA data. Despite the appealing simplicity and success of the GBW model it suffers from clear shortcomings. In particular it does not include scaling violations, that is, at large Q^2 it does not match with QCD (DGLAP) evolution. Therefore, Bartels, Golec-Biernat and Kowalski (BGBK) proposed a modification of the original ansatz of by replacing Q_s^2 by a gluon density with explicit DGLAP evolution:

$$\sigma_{q\bar{q}}^{BGBK}(x,r) = \sigma_0(1 - e^{\frac{-\pi^2 r^2 \alpha_s(\mu^2) x g(x,\mu^2)}{3\sigma_0}})$$

  The BGBK form of the dipole cross section led to significantly better fits to the HERA F_2 data than the original GBW model, especially in the region of larger Q^2. The good agreement of the original model with the DIS diffractive HERA data was also preserved. However, the contribution from charm quarks was omitted in the BGBK analysis.[1]

- **CGC Model:** The DGLAP evolution, which is generally used in the analysis of HERA data, may not be appropriate when x approaches the saturation region. Therefore, Iancu, Itakura and Munier proposed a new saturation model, the Colour Glass Condensate (CGC) model, in which gluon saturation effects are incorporated. CGC Dipole cross sections are[1]:

$$\sigma_{q\bar{q}}^{CGC}(x,r) = \sigma_0 \times \begin{cases} \mathcal{N}_0 \left(\frac{rQ_s}{2}\right)^{2(\gamma_s + \frac{1}{k\lambda Y} ln \frac{2}{rQ_s})} & rQ_s \leq 2 \\ 1 - e^{-Aln^2(BrQ_s)} & rQ_s > 2 \end{cases}$$

- **KT Model:** The CGC and FS models provide a description of HERA inclusive and diffractive DIS data which is better than the original GBW model and comparable in quality to the BGBK analysis. But all approaches to dipole saturation discussed so far ignored a possible impact parameter dependence of the dipole cross-section. This dependence was introduced in this context by KT, who assumed that the dipole cross-section is a function of the opacity Ω. Introduced impact parameter dependence to the cross-section and led to the b-CGC, b-Sat models development[1]:

$$\frac{d\sigma}{d^2\mathbf{b}} = 2\left(1 - \exp\frac{-\Omega}{2}\right)$$

$$\Omega = \frac{\pi^2}{N_c} r^2 \alpha_s(\mu^2) x g(x,\mu^2) T(b)$$

3 Project Motivation

Over the last two decades, the Color Dipole Picture has been developed to study high-energy scattering in QCD. An attractive feature of the dipole approach to high-energy interactions is that it gives a clear interpretation of the physics at small values of x. Even in its simplest form, it turns out to be rather successful in describing the total and diffractive virtual photon-nucleon cross-sections. Its generalizations are now commonly used for the parametrization of data from HERA [2].

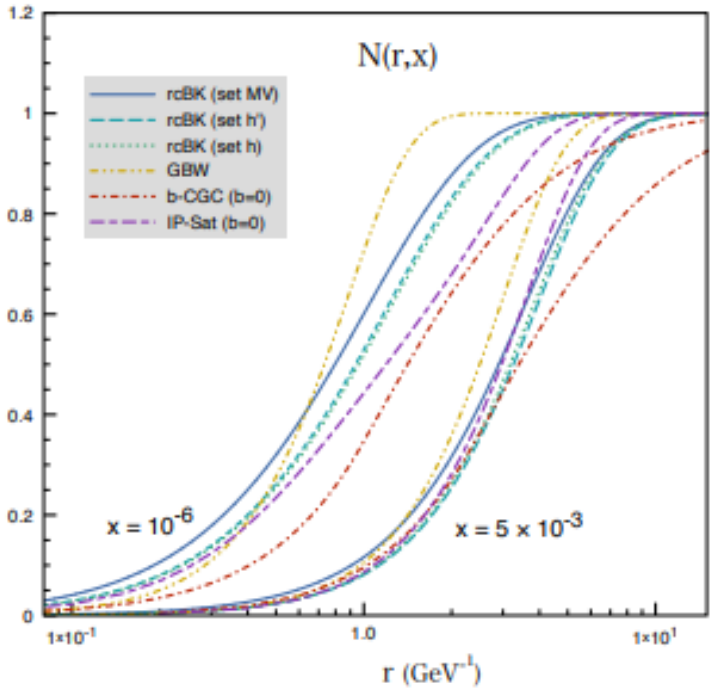


Figure 2: Dipole-proton scattering amplitude at two values of Bjorken-x: 5×10^{-3} and 10^{-6} [3]

Although these physics models are really powerful in describing the saturation condition at low x, we aim to investigate the possibility of a best-fitting Neural Network on the data by HERA to get an unbiased image of the process and compare the results of these models with our fit.

We have a good enough picture of how x dependence affects the dipole cross section, but our analysis will shed more light on how r (the dipole size) affects the dipole cross section of interaction. Further we can extend our technique to study about the impact of b (the impact parameter) on the cross section.

Once we have a fit on the HERA data, it will help in the development of more robust theoretical physics frameworks that explain the data more precisely.

The Inclusive DIS process is easy to detect in comparison to the exclusive DIS as we neglect the evolution of vector mesons and only focus on analyzing the energy and momentum transfer between the lepton and the nucleon. Due to this, there is much more data available for the Inclusive DIS compared to the Exclusive case and since we want a generalized picture of the interaction, we will be using our understanding of the Inclusive DIS to study the cross-section of interaction between the nucleon and lepton to understand the gluon and quark distribution.

4 Neural Networks

4.1 Introduction to Neural Networks

Machine learning is a field of artificial intelligence (AI) that focuses on the development of algorithms and models enabling computers to learn and improve from experience without being explicitly programmed. It involves creating systems that can automatically learn and make predictions or decisions based on data.
These algorithms and models learn patterns and relationships within the data, allowing computers to recognize and adapt to new information, make predictions, classify data, or optimize processes without human intervention. Machine learning techniques encompass various approaches like supervised learning, unsupervised learning, reinforcement learning, and deep learning, each suited for different types of tasks and data.

Neural networks, also known as artificial neural networks (ANNs) or simulated neural networks (SNNs), are a subset of machine learning and are at the heart of deep learning algorithms. Their name and structure are inspired by the human brain, mimicking the way that biological neurons signal to one another.

Artificial neural networks (ANNs) are comprised of a node layers, containing an input layer, one or more hidden layers, and an output layer. Each node, or artificial neuron, connects to another and has an associated weight and threshold. If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along to the next layer of the network.[7, 8, 9]

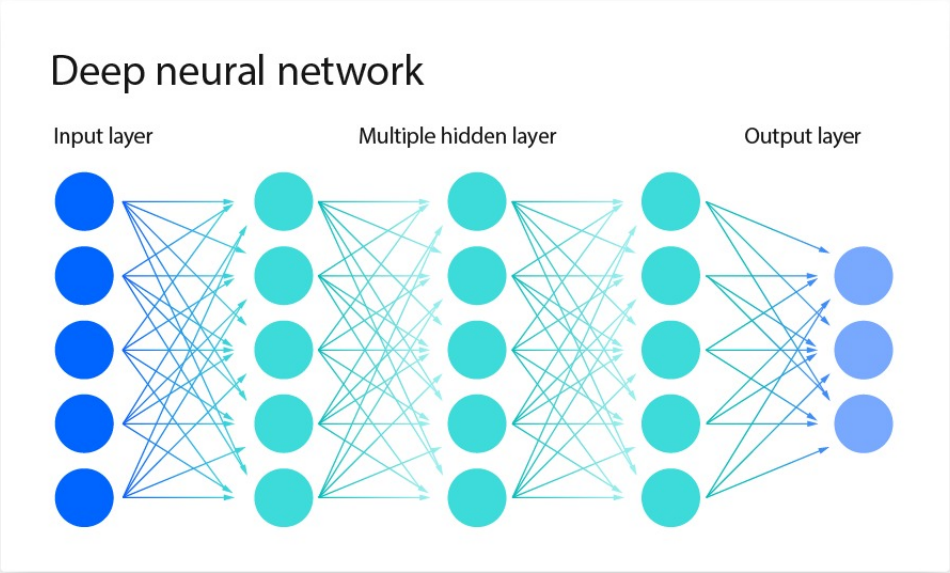


Figure 3: Structure of a Neural Network, replicating the Human Brain [7]

4.2 Components of Neural Networks

1. **Input Layer:** This layer receives the initial data or features. Each node in the input layer represents a feature of the input data.

2. **Hidden Layers:** These layers, which can be one or more, process the input data through weighted connections. Each connection has a weight that is adjusted during the training process to learn from the data.

3. **Weights and Biases:** Weights are parameters that the neural network learns during training. They determine the strength of connections between neurons. Biases are additional parameters that allow the model to learn the offset from zero.

4. **Activation Function:** Each neuron applies an activation function to the weighted sum of its inputs. Common activation functions include the sigmoid, hyperbolic tangent (tanh), and rectified linear unit (ReLU).

5. **Output Layer:** The final layer produces the output of the model. The number of nodes in this layer depends on the type of problem the neural network is designed to solve. For example, a binary classification task may have one output node, while a multi-class classification task may have multiple output nodes.

6. **Loss Function:** This is a measure of the difference between the predicted output and the actual target values. The goal during training is to minimize this loss.

7. **Optimization Algorithm:** The optimization algorithm adjusts the weights and biases in the network to minimize the loss function. Common optimization algorithms include stochastic gradient descent (SGD) and its variants.

4.3 Training of Neural Networks

1. **Forward Pass:** During the forward pass, input data is fed into the neural network. It passes through multiple layers of neurons, where each neuron applies a weighted sum of inputs followed by an activation function to produce an output.

2. **Calculating Error:** The output produced by the network is compared to the desired output (ground truth) to compute the error or loss function, which quantifies the difference between the predicted and actual values.

3. **Backward Pass (Backpropagation):** Backpropagation involves propagating this error backward through the network to update the weights of the neurons in order to minimize the error. It does this by applying the chain rule from calculus to compute the gradient of the loss function with respect to each weight in the network.

4. **Updating Weights:** The computed gradients indicate the direction and magnitude of adjustment required for each weight to reduce the error. The weights are updated in small steps (determined by a learning rate) in the direction opposite to the gradient, aiming to minimize the error gradually.

5. **Iterations:** This process of forward pass, error computation, backward pass, and weight updates iterates multiple times (epochs) over the entire dataset until the network's predictions improve, and the error is minimized to an acceptable level.

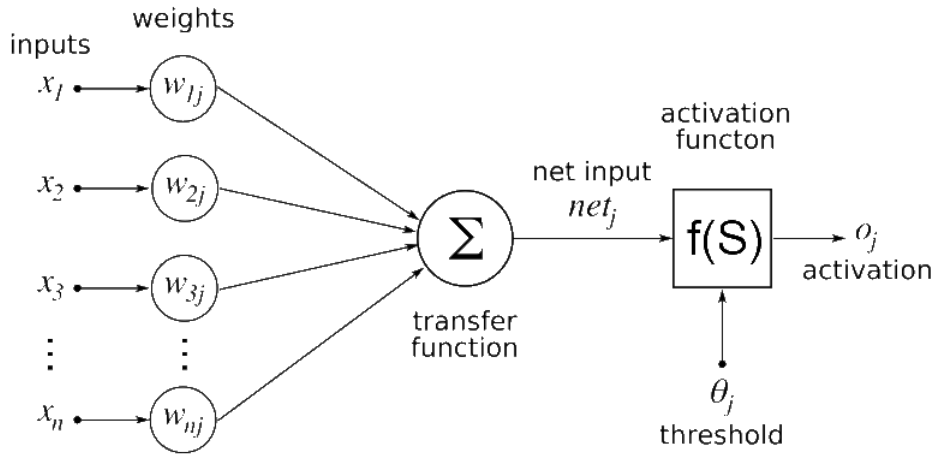


Figure 4: Working of a Perceptron algorithm, a single unit of Neural Network [8]

5 Computational Setup and Methodology

5.1 Cross-section calculations

The project will be done in Python using the **Pytorch** module for Machine Learning. Modules like **scipy**, **numpy** and motivation from **Sartre** is used to produce the Wave overlap functions, and the cross sections of the interaction.

The dipole cross section will be modelled as a Neural Network by training on data by HERA and then a fit will be obtained for the $\sigma_{T,L}^{\gamma^* p}$ cross-section. We will further obtain the reduced cross-section, to compare with the HERA data and calculate the loss to backpropagate in the network.

Multiple Neural Networks will be tested in course of multiple iterations to select the best hyper-parameters to obtain the best fit for our use case.

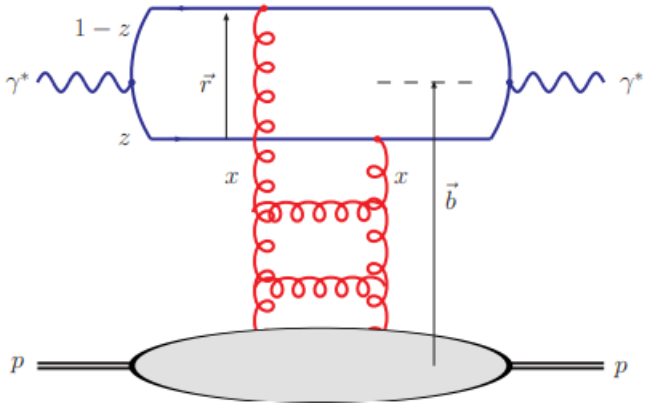


Figure 5: Elastic scattering amplitude for inclusive DIS [1]

The total cross-section for $\gamma^* p$ scattering is obtained by integrating the dipole cross section with the photon wave functions:

$$\sigma_{T,L}^{\gamma^* p}(x,Q) = Im\mathcal{A}_{q\bar{q}}(x,r,\Delta=0) = \sum_f \int d^2r \int_0^1 \frac{dz}{4\pi} (\Psi^*\Psi)_{T,L}^f \sigma_{q\bar{q}}(x,r)$$

$$\sigma_{T,L}^{\gamma^* p}(x,Q) = \sum_f \int d^2r \int_0^1 \frac{dz}{4\pi} (\Psi^*\Psi)_{T,L}^f \int d^2b \frac{d\sigma_{q\bar{q}}(x,r)}{d^2b}$$

Wave Overlap Equations:

$$(\Psi^*\Psi)_T^f = \frac{1}{2} \sum_{\substack{h,\bar{h}=\pm\frac{1}{2} \\ \lambda=\pm 1}} \Psi_{h\bar{h},\lambda}^* \Psi_{h\bar{h},\lambda} = \frac{2N_c}{\pi} \alpha_{em} e_f^2 \{[z^2 + (1-z)^2]\epsilon^2 K_1^2(\epsilon r) + m_f^2 K_0^2(\epsilon r)\}$$

$$(\Psi^*\Psi)_L^f = \frac{1}{2} \sum_{h,\bar{h}=\pm\frac{1}{2}} \Psi_{h\bar{h},\lambda=0}^* \Psi_{h\bar{h},\lambda=0} = \frac{8N_c}{\pi} \alpha_{em} e_f^2 Q^2 z^2 (1-z)^2 K_0^2(\epsilon r)$$

The proton structure functions $F_2(x,\, Q^2)$ and $F_L\ (x,\, Q^2)$ can be written in terms of the total photon-proton cross-sections as:

$$F_2(x,Q^2) = \frac{Q^2}{4\pi\alpha_{em}} (\sigma_T^{\gamma^* p} + \sigma_L^{\gamma^* p})$$

$$F_L(x, Q^2) = \frac{Q^2}{4\pi\alpha_{em}} \sigma_L^{\gamma^* p}$$

We will use the above F_2 and F_L which will further be obtained from our Neural Network outputs of $\sigma_T^{\gamma^* p}$ and $\sigma_L^{\gamma^* p}$. The F_2 and F_L will be used to obtain the reduced cross-section from the following equation, which we will use to compare the reduced cross-section from the data to calculate the loss of the neural network:

$$\sigma_r(x, y, Q^2) = F_2(x, Q^2) - \frac{y^2}{1 + (1 - y)^2} F_L(x, Q^2)$$

where $y = \frac{Q^2}{x \cdot s}$, and $\sqrt{s}$ is centre of mass energy

Variables description:

- r - Transverse size of Dipole
- Q^2 - Momentum transfer between scattered photon and scattered proton
- x - (Bjorken-x) Fraction of the momentum carried by a parton (quark or gluon) inside the proton along the direction of the incident photon
- z - Fraction of photon's momentum carried by the quark
- b - Impact Parameter, quantifies how "off-center" the collision is relative to the original trajectories of the particles
- f - Flavor (up, charm, down, bottom, top, and strange)
- $\mathcal{A}_{q\bar{q}}$ - Elementary amplitude for the scattering of a dipole of size r on the proton
- $\sigma_{q\bar{q}}$ - Total cross-section for the $q\bar{q}$ pair scattering on the proton (This will be our Neural Network)
- $\sigma_{T,L}^{\gamma^* p}$ - Total cross-section (transverse and longitudinal) for $\gamma^* p$ scattering
- $(\Psi^* \Psi)_{L,T}^{f}$ - Wave Overlap in transverse (T) and longitudinal (L) directions for a flavor f
- K_0, K_1 - Bessel's Function of the second kind and its derivative

5.2 Training the neural network

We used two methods to try training our Neural Network. Essentially we wanted to numerically integrate over r for a value of x, and then get the prediction of $\sigma_{q\bar{q}}$. Further, we would integrate over the impact parameter b as well, to study its dependence on the $\sigma_{q\bar{q}}$.

- **Method- 1:** In the first Neural Network architecture, we gave an input of x and multiple value of r for a particular x.

  Multiple such outputs were obtained for r_i and corresponding $\frac{d\sigma}{dr_i}$ were averaged over to get $\sigma_{q\bar{q}}$. The primary idea being that, if we summed over r_i for sufficiently large values such that r_i - r_{i+1} is very small, then we produce the effect of integration over r.

  After we have $\sigma_{q\bar{q}}$, we can obtain $\sigma_{T,L}^{\gamma^* p}$ from the equations in section 5.1 by just integrating over z and including the Wave overlap. Then we will find the proton structure functions F_2 and F_L which will give us the reduced cross-section σ_r to compute the loss.

Neural Network Architecture:

- Adam optimizer was used for backpropagation.
- Multiple Learning rates were tested $(10 - 10^{-5})$.
- No. of Hidden layers - 1
- Loss function - Mean Squared Error (MSE)

```
class NeuralNet(nn.Module):
  def __init__(self, inputs, outputs=1):
    super().__init__()
    self.inputLayer = nn.Linear(inputs, 2)
    self.hidden = nn.Linear(2, 2)
    self.relu = nn.ReLU()
    self.out = nn.Linear(2, outputs)

  def forward(self, x):
    pred = self.inputLayer(x)
    pred = self.hidden(pred)
    pred = self.relu(pred)
    pred = self.out(pred)
    return pred
```

Observations:

- Deep Inelastic Scattering data provided by the supervisor was used. No train-test split was performed as the data was very small (about 400 data points) and there were no signs of overfitting.
- For this model, we were unable to backpropagate due to the last step where we take the average corresponding to all r values (the optimizer cannot backpropagate due to this operation). Due to this no weight updates happen and model remains untrained. We were unable to figure out the right way to fix this issue. Hence we tried a different way to build the neural network, which is discussed ahead.

	sqrt(s)	Q2	x	sigma_r
1	318	0.85	0.0000124	0.552
2	300	0.85	0.0000138	0.614
3	300	0.85	0.00002	0.606
4	318	0.85	0.00002	0.611
5	318	0.85	0.0000398	0.558
6	318	0.85	0.00005	0.541
7	318	0.85	0.0001	0.503
8	318	0.85	0.000251	0.397
9	318	0.85	0.0008	0.349
10	318	0.85	0.0032	0.308

Figure 6: Snapshot of the data used

- **Method- 2:** Because the computation graph for the optimizer was breaking in the previous method due to simply taking the average of the $\frac{d\sigma}{dr_i}$, we considered taking a different approach.
- In this method, we used two neural networks stack, the first neural network stays the same as method- 1, where the output of ith neural network is $\frac{d\sigma}{dr_i}$. The second neural network works like the integral operation, instead of taking the average of all $\frac{d\sigma}{dr_i}$, it is learned by the model itself.
- The first part of the neural network takes the matrix of x vs multiple values of r as input and the second part takes the output of the above neural network and the Q^2 value as input to finally compute the reduced cross-section.

Neural Network Architecture:

- Adam optimizer was used for backpropagation.
- Learning rate - 10^{-3}.
- Loss function - Mean Squared Error (MSE)
- Epochs - 100

```
class ModifiedNN(nn.Module):
  def __init__(self, inputs, qdim, outputs):
    super().__init__()
    self.ip0 = nn.Linear(inputs, 5)
    self.h0 = nn.Linear(5, 5)
    self.op0 = nn.Linear(5, outputs)
    self.ip1 = nn.Linear(outputs+qdim, 3)
    self.h1 = nn.Linear(3, 3)
    self.op1 = nn.Linear(3, 1)

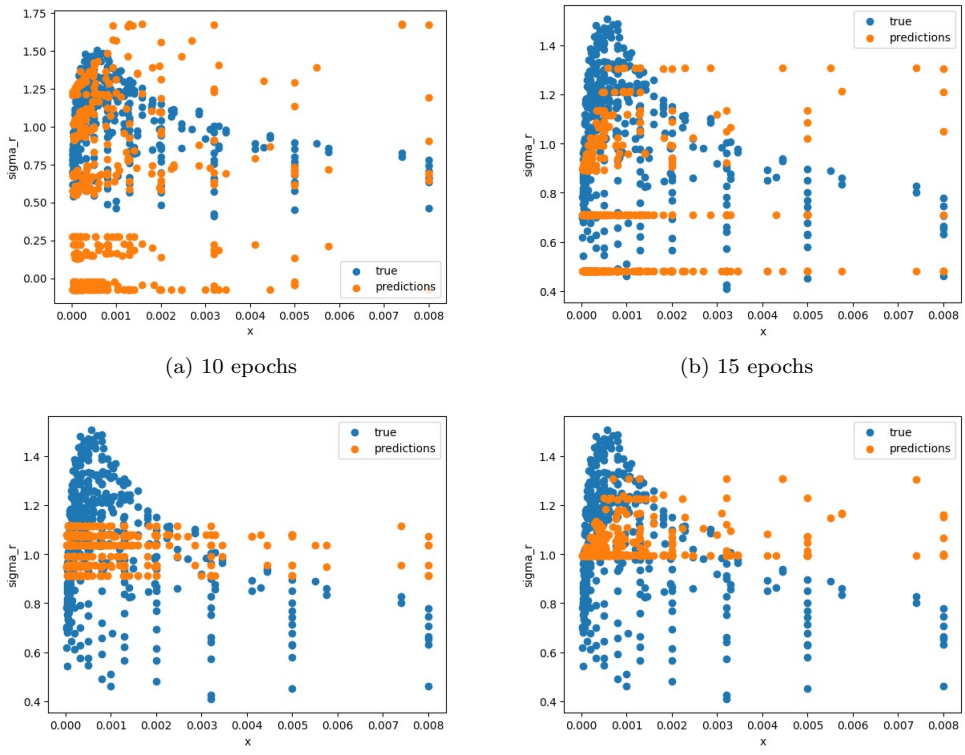
  def forward(self, x, q2):
    a, b = x.shape[1], x.shape[2]
    x = x.view(-1, a*b)
    z = self.ip0(x)
    z = torch.relu(z)
    z = self.h0(z)
    z = nn.Dropout(0.4)(z)
    z = torch.relu(z)
    z = self.op0(z)
    y = torch.cat((z, q2), dim=1)
    y = self.ip1(y)
    y = nn.Dropout(0.4)(y)
    y = torch.relu(y)
    y = self.h1(y)
    y = nn.Dropout(0.4)(y)
    y = torch.relu(y)
    y = self.op1(y)
    return z, y
```

Observations:

- This model backpropagates successfully, with weights of different neurons being updated by the optimizer, normally.
- The issue with this approach is that we can, in no way, get cross-section information for single values of r as the model expects us to give a matrix that consists of multiple values of r. We cannot change the input to the neural network.
- We tried to pull this information (cross-section values for a single r) by masking all other inputs and only activating one input, but the results were not interpretable.
- This was the major shortcoming of this approach, although we have a working neural network, we cannot analyze or interpret the results effectively to make much sense.
- The model was also significantly undertrained, a possible reason could be smaller dataset.

5.3 Further avenues to explore

- The results with current models were not interpretable or satisfactory. We aim to explore different architecture, which could help us study the r dependence of the reduced cross-section as well explore ways to interpret the results of the current model.
- Experimenting with different and more data would be helpful. The problem of underfitting can be resolved with this.



(a) 10 epochs

(b) 15 epochs

(c) 100 epochs

(d) 500 epochs

Figure 7: Bjorken-x vs Reduced cross sections for the fit at different epochs

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